

Rules of Probability

Tennis Tournament Example

Suppose that n tennis players are entered in a tournament. In the first round, the players are paired one against another at random. The loser in each pair is eliminated from the tournament, and the winner in each pair continues into the second round. If the number of players n is odd, then one player is chosen at random before the pairings are made for the first round, and that player automatically continues into the second round. All the players in the second round are then paired at random. Again, the loser in each pair is eliminated, and the winner in each pair continues into the third round. If the number of players in the second round is odd, then one of these players is chosen at random before the others are paired, and that player automatically continues into the third round. The tournament continues in this way until only two players remain in the final round. They then play against each other, and the winner of this match is the winner of the tournament. We shall assume that all n players have equal ability. Find the probability that two specific players A and B will ever play against each other during the tournament.

Tennis Tournament Solution

- We shall first determine the total number of matches that will be played during the tournament. After each match has been played, one player—the loser of that match—is eliminated from the tournament. The tournament ends when everyone has been eliminated from the tournament except the winner of the final match. Since exactly $n - 1$ players must be eliminated, it follows that exactly $n - 1$ matches must be played during the tournament.
- The number of possible pairs of players is $\binom{n}{2}$. Each of the two players in every match is equally likely to win that match, and all initial pairings are made in a random manner. Therefore, before the tournament begins, every possible pair of players is equally likely to appear in each one of the $n - 1$ matches to be played during the tournament. Accordingly, the probability that players A and B will meet in some match that is specified in advance is $1/\binom{n}{2}$. If A and B do meet in that particular match, one of them will lose and be eliminated. Therefore, these same two players cannot meet in more than one match.
- It follows from the preceding explanation that the probability p that players A and B will meet at some time during the tournament is

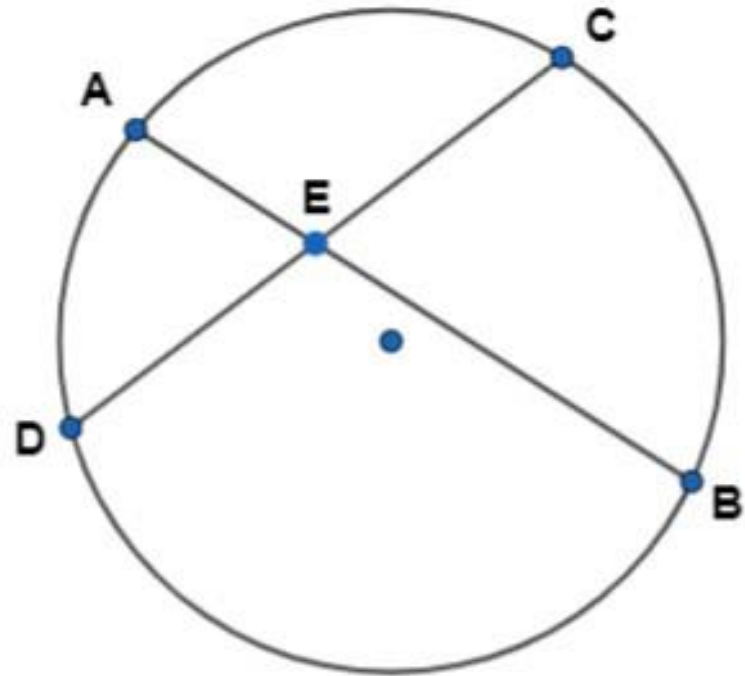
$$p = \frac{n - 1}{\binom{n}{2}} = \frac{2}{n}$$

Data Science interview question

What is the probability that two random chords AB and CD intersect at a point inside the circle?

Solution:

- The first point, A, can be anywhere on the circle without affecting the result.
- Starting from A, and going clockwise we must place the remaining 3 points B, C and D.
- There are $3! = 6$ ways to do that.
- The chords will cross inside the circle only when B is between C and D, which are the orderings CBD and DBC.
- Therefore, the probability of the chords crossing is $2/6 = 1/3$



Proof of Compliment of Event

For any event E , we can write:

$$S = E \cup \bar{E} \text{ \& } E \cap \bar{E} = \emptyset$$

hence

$$1 = P[S] = P[E \cup \bar{E}] = P[E] + P[\bar{E}]$$

or

$$P[\bar{E}] = 1 - P[E]$$

Note: $\bar{\bar{S}} = \phi$

$$\text{Hence } P[\phi] = P[\bar{\bar{S}}] = 1 - P[S] = 1 - 1 = 0$$

Example (Lotto 6/49): Select six numbers from 1 to 49

a) How many possible different tickets are there?

Answer:

$$\begin{aligned}\binom{49}{6} &= {}_{49}C_6 = \frac{49!}{6!43!} = \frac{49(48)(47)(46)(45)(44)}{6(5)(4)(3)(2)(1)} \\ &= 13,983,816\end{aligned}$$

b) How many combinations have 0 winning numbers?

c) How many combinations have 1 winning number?

d) How many combinations have 2 winning numbers?

e) What is the probability of not winning anything?

Exercise

A city with 6 districts has 6 robberies in a particular week. Assume the robberies are located randomly, with all possibilities for which robbery occurred where equally likely.

What is the probability that some district had more than 1 robbery?

Example 1

What is the exact probability that a *specific* bin receives exactly k balls when m distinct balls are thrown into n distinct bins?

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Solution:

Each of the m distinct balls can be placed into any of the n distinct bins. The total number of ways to distribute all m balls is:

$$n \times n \times \cdots \times n = n^m$$

We want one specific bin to contain exactly k balls. To count how many ways this can happen, we break it into two choices:

Choose the balls: Select *which* k balls out of the total m balls will go into the specific bin: $\binom{m}{k}$

Place the remaining balls: The remaining $m - k$ balls must be placed into the other $n - 1$ bins (to ensure the specific bin gets *exactly* k balls and no more).

Each of these $m - k$ remaining balls has $n - 1$ valid destination bins. The number of ways to distribute them is: $(n - 1)^{m-k}$

Multiplying these independent choices gives the total number of favorable arrangements:

$$\text{Favorable Outcomes} = \binom{m}{k} (n - 1)^{m-k}$$

Example 1

What is the exact probability that a *specific* bin receives exactly k balls when m distinct balls are thrown into n distinct bins?

Solution:

The probability is the ratio of favorable outcomes to total possible outcomes:

$$P = \frac{\binom{m}{k} (n-1)^{m-k}}{n^m}$$

This fraction is the purely counting-based derivation of the probability. By splitting the denominator n^m into $n^k \cdot n^{m-k}$, it algebraically rearranges directly into the standard binomial format:

$$P = \binom{m}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{m-k}$$

Example 2: The birthday problem

In a room of n individuals, what is the probability that at least 2 have the same birthday. (Assuming 365 days)

Solution:

Let E = the event that at least 2 have the same birthday.

Then $\bar{E}$ = the event no pair of individuals have the same birthday.

We will find $P(E)$ as $P(E) = 1 - P(\bar{E})$

$n(S)$ = total number of ways of assigning the n individuals birthdays = 365^n (why?)

$n(\bar{E})$ = total number of ways of assigning the n distinct birthdays

Hence,

$$P(\bar{E}) = \frac{n(\bar{E})}{n(S)} = \frac{P_{365,n}}{365^n}$$

If your calculator has only combinations and factorial.

$$P[E] = 1 - P[\bar{E}] = 1 - \frac{\binom{365}{n} n!}{365^n}$$

$$= 1 - \frac{\frac{365!}{n!(365-n)!} n!}{365^n} = 1 - \frac{365!}{(365-n)! 365^n}$$

$$= 1 - \left(\frac{365}{365}\right) \left(\frac{364}{365}\right) \left(\frac{363}{365}\right) \cdots \left(\frac{365-n+1}{365}\right)$$

Thus $P\left[\begin{array}{c} \text{Two people have same birthday} \\ \text{in a room of } n \text{ people} \end{array}\right]$

$$= 1 - \left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right) \cdots \left(\frac{365 - n + 1}{365}\right)$$

Table:

n	$P[E]$
10	0.1169
20	0.4114
22	0.4757
23	0.5073
30	0.7063
40	0.8912

← 50%

Counting Permutations When the Objects Are Not All Distinct

Theorem: The number of ways to arrange n objects, of which n_1 being of one kind, n_2 of a second kind, $\dots$, and n_r of r^{th} kind is:

$$\frac{n!}{n_1! n_2! \dots n_r!}$$

where we have that $n_1 + \dots + n_r = n$.

Example 3:

Two monkeys, Mickey and Marian, are strolling along a moonlit beach when Mickey sees an abandoned Scrabble set. Investigating, he notices that some of the letters are missing, and what remain are the following fifty-nine:

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>	<i>M</i>
4	1	2	2	7	1	1	3	5	0	3	5	1

<i>N</i>	<i>O</i>	<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
3	2	0	0	2	8	4	2	0	1	0	2	0

Mickey, being of a romantic bent, would like to impress Marian, so he rearranges the letters in hopes of spelling out something clever. (*Note:* The rearranging is random because Mickey can't spell; fortunately, Marian can't read, so it really doesn't matter.) What is the probability that Mickey gets lucky and spells out

She walks in beauty, like the night
Of cloudless climes and starry skies

Example (Solution):

As we might imagine, Mickey would have to get *very* lucky. The total number of ways to permute fifty-nine letters—four *A*'s, one *B*, two *C*'s, and so on—is a direct application of Theorem 2.6.2:

$$\frac{59!}{4!1!2! \dots 2!0!}$$

But of that number of ways, only one is the couplet he is hoping for. So, since he is arranging the letters randomly, making all permutations equally likely, the probability of his spelling out Byron's lines is

$$\frac{1}{\frac{59!}{4!1!2! \dots 2!0!}}$$

or, using Stirling's formula, about 1.7×10^{-61} . Love may conquer all, but it won't beat those odds: Mickey would be well advised to start working on Plan B. ■

Additive rule (two events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

And simplifies to:

$$P(A \cup B) = P(A) + P(B) \text{ if } A \cap B = \phi$$

Proof:

We can express the union of A and B , denoted as $A \cup B$, as the combination of three non-overlapping, mutually exclusive regions:

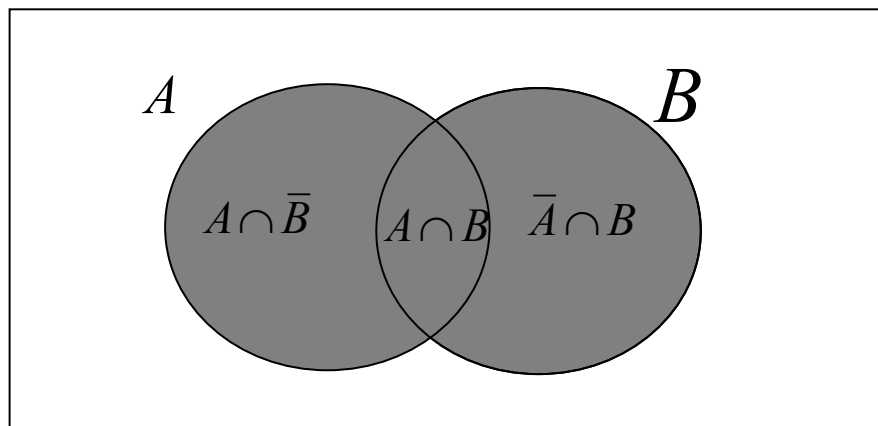
Only A occurs: $A \cap B^c$ (where B^c is the complement of B)

Only B occurs: $B \cap A^c$

Both occur: $A \cap B$

Because these three regions do not overlap, by Axiom 3, the probability of $A \cup B$ is the simple sum of their probabilities:

$$P(A \cup B) = P(A \cap B^c) + P(B \cap A^c) + P(A \cap B)$$



But,

$$A = (A \cap B) \cup (A \cap \bar{B})$$

and the two events in the union are disjoint.

Consequently,

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap \bar{B}) \\ \Rightarrow P(A \cap \bar{B}) &= P(A) - P(A \cap B) \end{aligned}$$

Similarly,

$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$\Rightarrow P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Substitute in the equation from previous slide to obtain:

$$\begin{aligned} P(A \cup B) &= P(A \cap \bar{B}) + P(A \cap B) + P(\bar{A} \cap B) \\ &= P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

Example 4

City A and city B are two of the cities competing for the World Olympic games. (There are also many others). The organizers are narrowing the competition to the **final 5 cities**.

There is a 20% chance that city A will be amongst the **final 5**. There is a 35% chance that city B will be amongst the **final 5** and an 8% chance that *both* city A and city B will be amongst the **final 5**. What is the probability that city A *or* city B will be amongst the **final 5**?

Hence

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

Solution:

Let A = the event that city A is amongst the **final 5**.

Let B = the event that city B is amongst the **final 5**.

Given $P[A] = 0.20$, $P[B] = 0.35$, and $P[A \cap B] = 0.08$

What is $P[A \cup B]$?

Note: “both” $\equiv \cap$, “or” $\equiv \cup$.

$$\begin{aligned} P[A \cup B] &= P[A] + P[B] - P[A \cap B] \\ &= 0.20 + 0.35 - 0.08 = 0.47 \end{aligned}$$

Another Example – *Bridge Hands*

A bridge hand is 13 cards chosen from a deck of 52 cards.

Total number of Bridge Hands

$$\begin{aligned} N = n(S) &= \binom{52}{13} = \frac{52(51)(50)(49)\text{L } (42)(41)(40)}{13(12)(11)(10)\text{L } (3)(2)(1)} \\ &= 635,013,559,600 \end{aligned}$$

Some Questions

1. What is the probability that the hand contains exactly 5 spades (8 non-spades)?
2. What is the probability that the hand contains exactly 5 hearts?
3. What is the probability that the hand contains exactly 5 spades **and** 5 hearts?
4. What is the probability that the hand contains exactly 5 spades **or** 5 hearts?
(i. e. a five card major)

Solutions

1. What is the probability that the hand contains exactly 5 spades (8 non-spades)?

Let A = the event that the hand contains exactly 5 spades (8 non-spades)?

$n(A)$ = the number of hands that contain exactly 5 spades (8 non-spades)?

$$= \binom{13}{5} \binom{39}{8} = (1,287)(61,523,748) = 79,181,063,676$$

spades ← *non-spades*

hand

the number of ways of choosing the 8 non-spades

the number of ways of choosing the 5 spades

$$\Rightarrow P(A) = \frac{n(A)}{n(S)} = \frac{79,181,063,676}{635,013,559,600} = 0.12469$$

2. What is the probability that a bridge hand contains exactly five 

Let B = the event that the hand contains exactly 5 hearts

Then

$$\begin{aligned} P(B) &= \frac{n(B)}{n(S)} = \frac{\binom{13}{5} \binom{39}{8}}{\binom{52}{13}} \\ &= \frac{79,181,063,676}{635,013,559,600} = 0.1246919 \end{aligned}$$

3. What is the probability that the hand contains exactly 5 spades **and** 5 hearts?

$A \cap B$ = the event that the hand contains exactly 5 spades **and** 5 hearts?

$$n(A \cap B) = \binom{13}{5} \binom{13}{5} \binom{26}{3}$$

the number of ways of
choosing the 3 non-
spades, hearts

the number of ways of
choosing the 5 spades

the number of ways of
choosing the 5 hearts

$$\begin{aligned} &= (1,287)(1,287)(61,523,748) \\ &= 4,306,559,400 \end{aligned}$$

$$\text{and } P[A \cap B] = \frac{4,306,559,400}{635,013,559,600} = 0.006781838$$

4. What is the probability that the hand contains exactly 5 spades **or** 5 hearts?
(i. e. a five-card major)

$$\begin{aligned}P[A \cup B] &= P[A] + P[B] - P[A \cap B] \\&= 0.124691926 + 0.124691926 - 0.006781838 \\&= 0.2426\end{aligned}$$

Thus there is a 24.26% chance that a bridge hand will contain at least one five card major.

Additive for Three Events

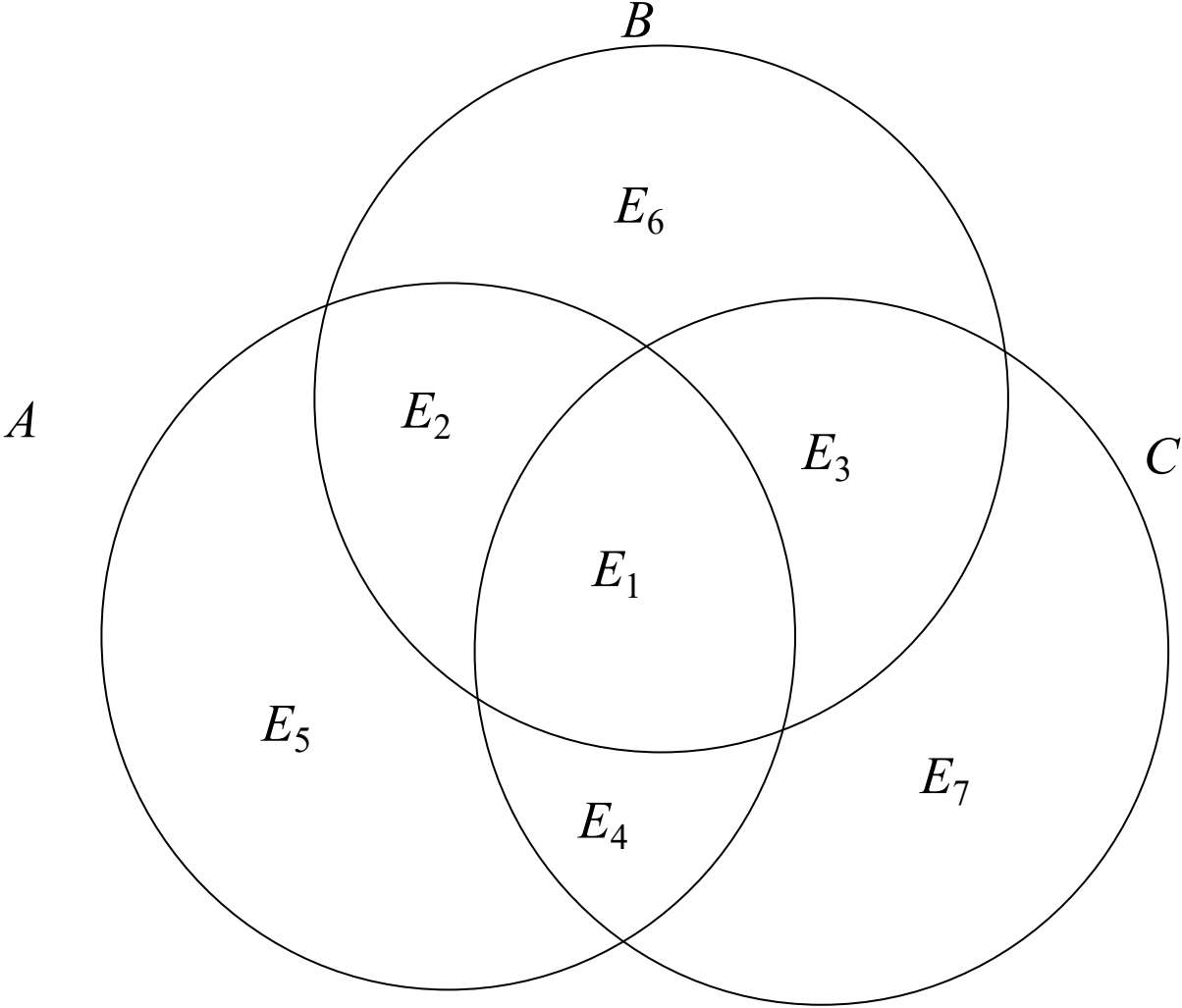
$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C) \end{aligned}$$

Special case:

if $A \cap B = \phi$, $A \cap C = \phi$ and $B \cap C = \phi$, then

$$P[A \cup B \cup C] = P[A] + P[B] + P[C]$$

Union of Three Events



$$A \cup B \cup C = E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5 \cup E_5 \cup E_7$$

$$A = E_1 \cup E_2 \cup E_4 \cup E_5$$

$$B = E_1 \cup E_2 \cup E_3 \cup E_6$$

$$C = E_1 \cup E_3 \cup E_4 \cup E_7$$

When these three are added
 E_2 , E_3 and E_4 are counted
 twice and E_1 is counted three
 times

$$A \cap B = E_1 \cup E_2$$

$$A \cap C = E_1 \cup E_4$$

$$B \cap C = E_1 \cup E_3$$

When these three are subtracted off
 the extra contributions of E_2 , E_3 and
 E_4 are removed and the contribution
 of E_1 is completely removed

$$A \cap B \cap C = E_1$$

To correct we now have to add back
 in the contribution of E_1 .

Example

Three fair dice are rolled. Find the probability that at least one die shows a 6.

Solution:

Let $A = \{\text{first die shows 6}\}$, $B = \{\text{second die shows 6}\}$, $C = \{\text{third die shows 6}\}$

Then

$$\begin{aligned} & P(A \cup B \cup C) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & \quad + P(A \cap B \cap C) \\ &= \frac{6^2}{6^3} + \frac{6^2}{6^3} + \frac{6^2}{6^3} - \frac{6}{6^3} - \frac{6}{6^3} - \frac{6}{6^3} + \frac{1}{6^3} = \frac{91}{216} = 0.4212963 \end{aligned}$$

Time to think: Is there an alternative solution?

Exercise

- Ten standard fair 6-sided dice are rolled simultaneously.
- What is the exact probability that **every** face (1 through 6) appears at least once?

Example

- a) Find the probability that a randomly selected integer n , in the range $1 \leq n \leq 2019$, is divisible by 2 or 3 or 7.

Answer:

$1442/2019$

- b) Find the probability that random seven-digit string contains at least five consecutive equal digits.

Answer:

$2800/(10^7)$

Rule (Inclusion-exclusion)

For any n events $A_1, \dots, A_n$,

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) \\ &\quad + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots \\ &\quad + (-1)^{n+1} P(A_1 \cap \dots \cap A_n) \end{aligned}$$

The formula simplifies if $A_i \cap A_j = \phi$ for all $i \neq j$.

then

$$P\left[\bigcup_{i=1}^n A_i\right] = \sum_{i=1}^n P[A_i]$$

Example

Suppose that we have a family of n people.

At a party they decide to put their names in a hat.

Each person will randomly pick a name.

This will be the only person for which they will buy a present.

Questions

- a) What is the probability that at least one person picks their own name?
- b) How does this probability change as the size of the family, n , increase to infinity?

Exercises

- Consider a set of n people drawing names from a hat. What is the probability that a specific subset of m people (where $m \leq n$) all draw names *other* than their own, while the remaining $n - m$ people can draw anyone's name?
- A norepeatword is a sequence of at least one (and possibly all) of the usual 26 letters a,b,c,...,z, with repetitions not allowed. For example, “course” is a norepeatword, but “statistics” is not. Order matters, e.g., “course” is not the same as “source”. A norepeatword is chosen randomly, with all norepeatwords equally likely. Show that the probability that it uses all 26 letters is very close to $1/e$.